

Multiple Choice (10 marks)

1. In Python 3, what is `['a', 'Chemistry', 0, 1][-3]`?
 - (a) a
 - (b) Chemistry
 - (c) 0
 - (d) 1
2. Which of the following spreadsheet functions would be most useful for detecting im-probably high or low values that have become part of a data set as a result of data entry errors?
 - (a) A function that averages numeric values in a column or row
 - (b) A function that counts the values in a column or row
 - (c) A function that rounds a numeric value
 - (d) A function that sorts values in a column or row
3. In Python 3, which of the following function convert a string to an int in python?
 - (a) `int(x [,base])`
 - (b) `long(x [,base] )`
 - (c) `float(x)`
 - (d) `str(x)`
4. What is the output of `"abc"[-1]` in Python 3?
 - (a) Error
 - (b) a
 - (c) b
 - (d) c
5. If a, b, and c are integers, which of the following conditions is sufficient to guarantee that the expression `a < c || a < b && !(a == c)` evaluates to true?
 - (a) `a < c`
 - (b) `a < b`
 - (c) `a > b`
 - (d) `a == b`

True / False (10 marks)

Answer True or False

6. True or False: In Python 3, the function `max(l)` where `l = [1,2,3,4]` returns the value 3.
7. True or False: An abstract syntax tree (AST) is responsible for managing information about variables and their attributes in a compiler.
8. True or False: The user's public key used for encryption poses the greatest threat to personal privacy online.
9. True or False: The values of local variables are typically not stored within a subroutine's activation record frame.
10. True or False: In Python 3, the expression `['a', 'Chemistry', 0, 1][-3]` evaluates to 'Chemistry'.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to list out the list of given strings individually using map function.
12. Write a function to re-arrange the elements of the given array so that all negative elements appear before positive ones.

End of Paper